

STAT 408 Lab 1: Data Viz with ggplot

Hadley Stabnau

Jamnuamry 15, 2026

1. Load the three packages you are using for this lab on the `penguins` dataset.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
```

```
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

`penguins`, `penguins_raw`

```
library(ggthemes)
```

Warning: package 'ggthemes' was built under R version 4.5.2

2. How many rows (observations) are in `penguins`? How many columns (variables)? What are the variables (column names) in this dataset? Answer these three questions using code covered in `lab01_notes.qmd`.

```
nrow(penguins)
```

```
[1] 344
```

```
ncol(penguins)
```

```
[1] 8
```

There are 344 observations (rows) in the penguin data set and 8 variables (columns).

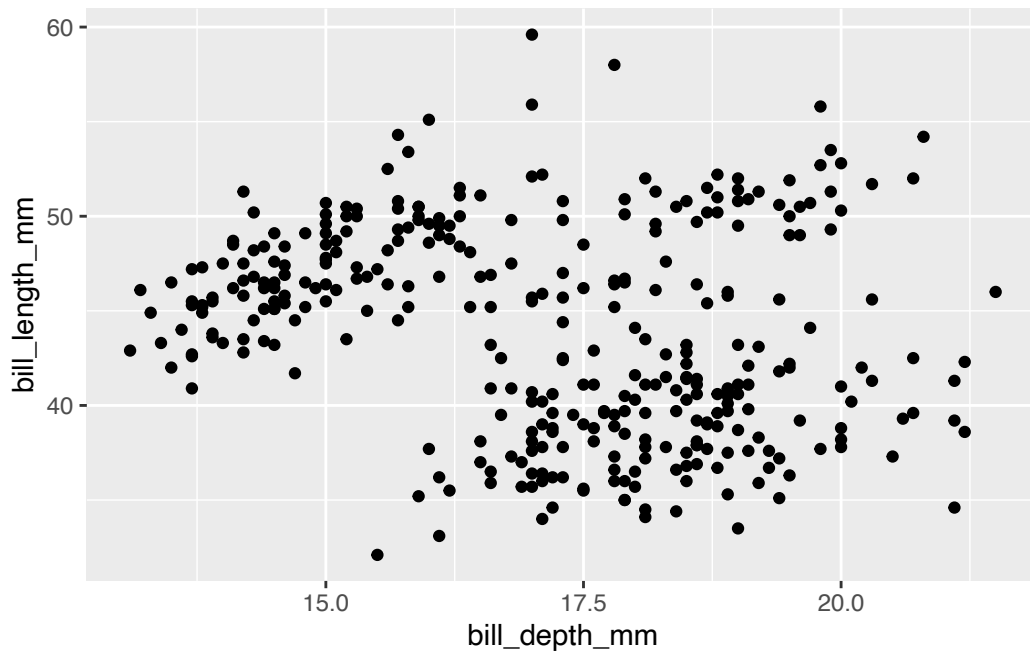
3. What does the `bill_depth_mm` variable in the `penguins` data frame describe? Read the help for `?penguins` to find out.

The `bill_depth_mm` variable in this data frame describes the bill depth in mm of a given penguin.

4. Make a scatterplot of `bill_depth_mm` vs. `bill_length_mm`. That is, make a scatterplot with `bill_depth_mm` on the y-axis and `bill_length_mm` on the x-axis. Describe (in text, not code or code comments) the relationship between these two variables.

```
# Code chunk 1
ggplot(
  data = penguins,
  mapping = aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point()
```

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).

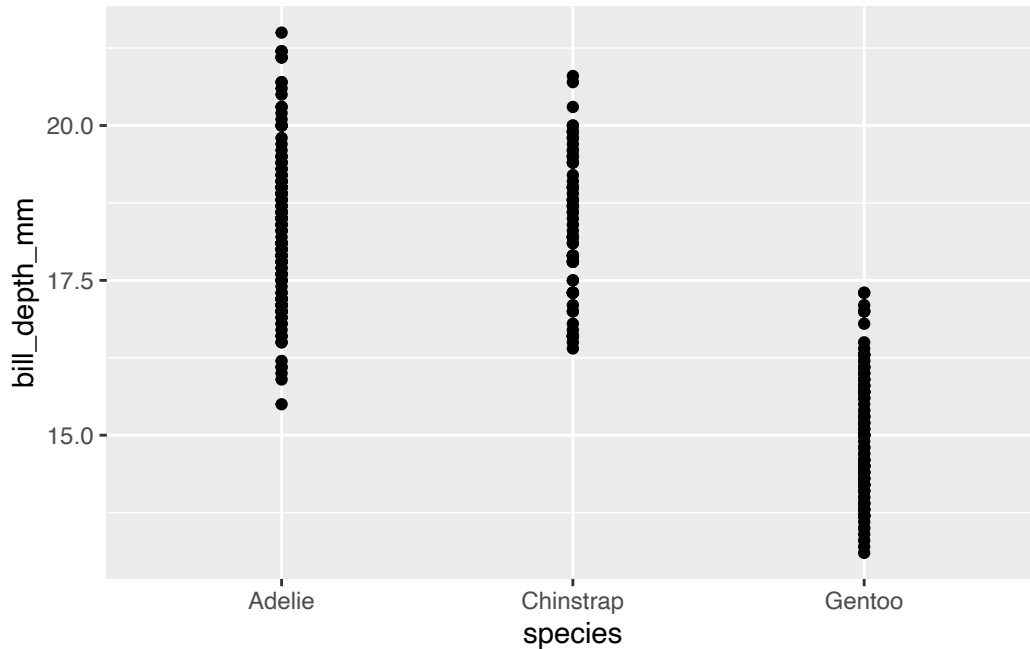


These two variables, bill depth and bill length, appear to have a weak positive linear relationship with two possible different groups. There do not appear to be any outliers in this data plot.

5. What happens if you make a scatterplot of `species` vs. `bill_depth_mm`? (Make and show this scatterplot, then discuss in text what happens.) What might be a better choice of geom for this plot? (See other options mentioned in Section 1.2.3.)

```
# Code chunk 1
ggplot(
  data = penguins,
  mapping = aes(x = species, y = bill_depth_mm)
) +
  geom_point()
```

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).



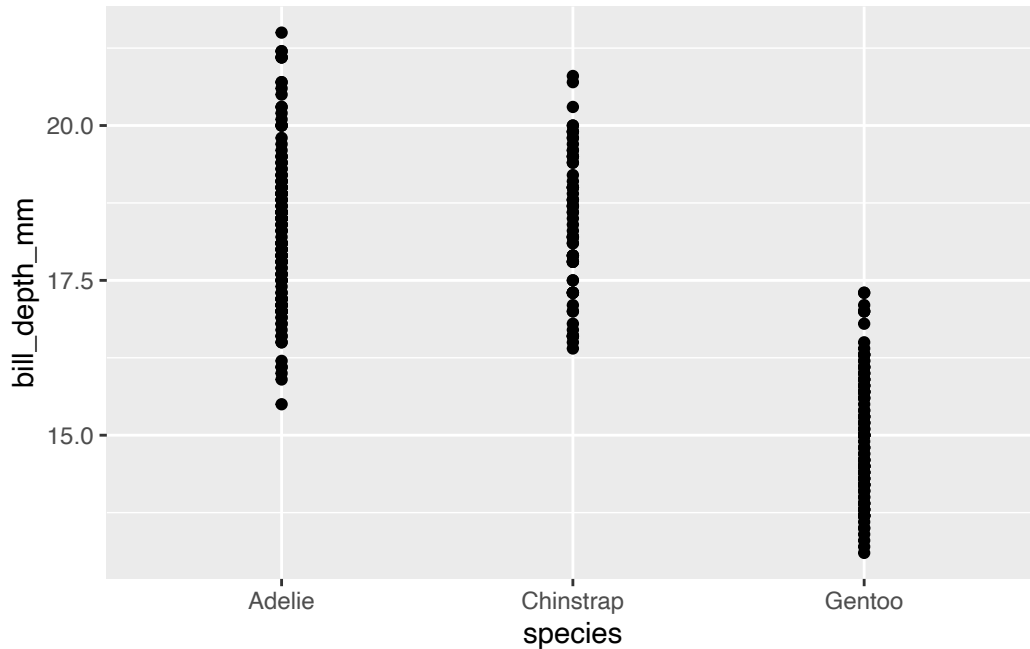
A better choice of geom for this plotted data would be some kind of bar graph. This is because each species will lie in the same bar/line anyways, and a bar graph would show the differences in bill depths by the height of each bar.

- Why does the following code give an error? (Write your answer in text.) Fix it. (NOTE: you need to delete the line that says `#| eval: false` so that this code will run when you render your Quarto document.)

The error in this chunk of code is that there is no variable assigned to the x or the y axis', and there is not any specific mapping assigned so R has the correct data but doesn't know what we want it to do with that data. To fix this, I just stole the variables that were used in the previous question.

```
ggplot(data = penguins, mapping = aes(x = species, y = bill_depth_mm)) +
  geom_point()
```

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).

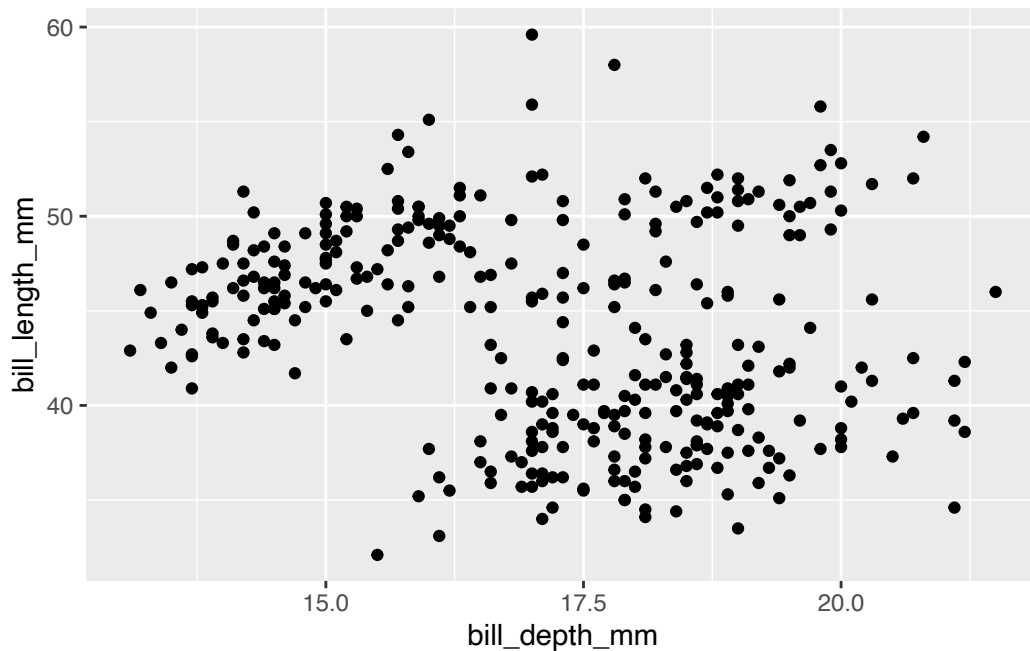


7. What does the `na.rm` argument do in `geom_point()`? What is the default value of the argument? (Hint: to answer these questions, look up the help page for `geom_point()` the same way you did for the `penguins` dataset, and don't include the parentheses when you do that.) Then create a scatterplot where you successfully use this argument set to `TRUE`.

The `na.rm` argument in `geom_point()` removes missing values, whether or not we set `na.rm` equal to `true` or `false` determines if the values are removed silently or with a warning message

```
ggplot(
  data = penguins,
  mapping = aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm=FALSE)
```

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).



8. Recreate the visualization in Exercise 8 of Section 1.2.5. What aesthetic should `bill_depth_mm` be mapped to? And should it be mapped at the global level or at the geom level?

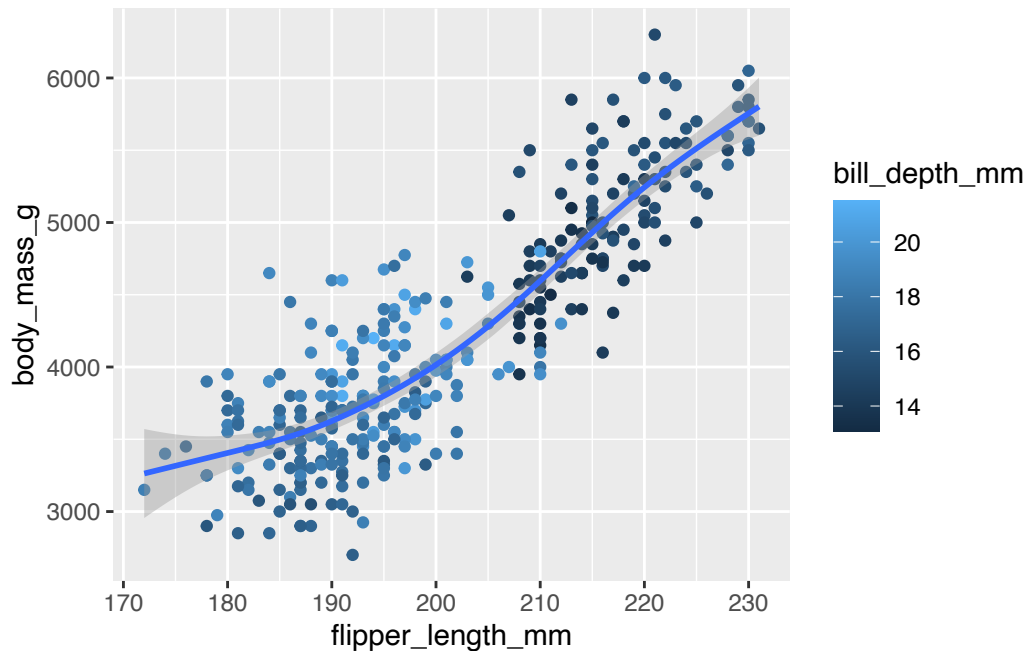
Bill depth will be mapped to the color aesthetic. This is at the geom level not the global level.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(mapping = aes(color = bill_depth_mm)) +
  geom_smooth(method = "gam")
```

```
`geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



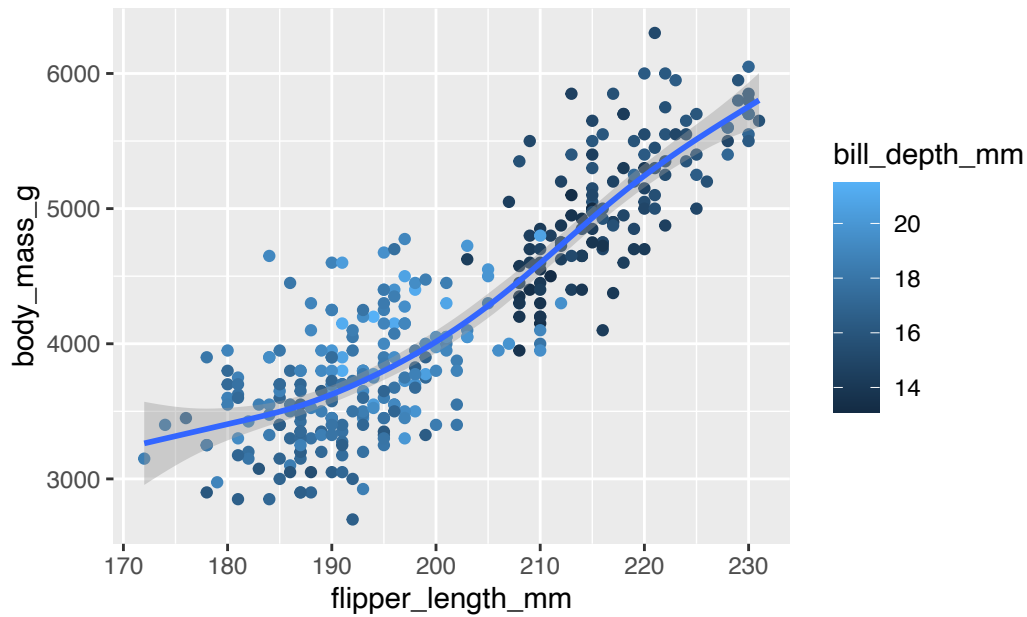
9. Add the following caption to the plot you made in the previous exercise: “Data come from the palmerpenguins package.” Hint: Take a look at the documentation for `labs()`.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(mapping = aes(color = bill_depth_mm)) +
  geom_smooth(method = "gam") +
  labs(caption="Data come from the palmerpenguins package")
```

```
`geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
```

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom\_point()`).



Data come from the palmerpenguins package

10. Will these two graphs look different? Why/why not? (Please leave the `#| eval: false` option in this code chunk.)

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point() +
  geom_smooth()

ggplot() +
  geom_point(
    data = penguins,
    mapping = aes(x = flipper_length_mm, y = body_mass_g)
  ) +
  geom_smooth(
    data = penguins,
    mapping = aes(x = flipper_length_mm, y = body_mass_g)
  )
```

These two graphs will look the same because even though their code is not identical, in the first plot the data is set globally which automatically sort of carries down to the geom point/smooth. Whereas in the second plot, the data is directly set inside of the geom point and geom smooth.

So overall, both of those code chunks are building the same plots, but just applying the data in different ways within the code.

11. Type the name of the function we used in this lab for each of the following tasks.
 - Getting the number of variables in a dataset: *ncol()*
 - Setting up plot aesthetics: *mapping=aes()*
 - Loading a package: *library()*
 - Getting the number of rows in a dataset: *nrow()*
 - Installing a package: *install.packages()*
12. Make a few notes here for yourself about some useful tips and tricks you learned (or were reminded of) in today's lab.

The tip I learned/reminded myself of today that was incredibly helpful was the `read in help` function, being able to just pull up information on my screen on pretty much anything in R is awesome. The `Label` function is also a new function I learned today that will certainly come in handy.